

GENERAL SCIENCE

Physics | Chemistry | Biology

For SSC | UPSC | Railway | Banking | State PSC | Defence Exams

PHYSICS - 1. UNITS & MEASUREMENTS

- > SI System: 7 base units - Metre (length), Kilogram (mass), Second (time), Ampere (current), Kelvin (temperature), Mole (amount), Candela (luminous intensity).
- > CGS System: Centimetre, Gram, Second.
- > Light Year: Distance light travels in one year ($\sim 9.46 \times 10^{12}$ km). Used for astronomical distances.
- > Parsec: 3.26 light years. Astronomical Unit (AU): Earth-Sun distance (~ 150 million km).
- > Angstrom: 10^{-10} metre. Used for wavelength of light.
- > Nautical Mile: 1852 metres. Used in navigation.

PHYSICS - 2. MOTION & FORCE

- > Newton's 1st Law (Inertia): Object at rest stays at rest; object in motion stays in motion unless acted upon by external force.
- > Newton's 2nd Law: $F = ma$ (Force = mass x acceleration).
- > Newton's 3rd Law: Every action has equal and opposite reaction.
- > Friction: Force that opposes relative motion. Static > Kinetic > Rolling friction.
- > Centripetal Force: Force directed towards center of circular path.
- > Gravitational Force: $F = Gm_1m_2/r^2$. $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$.
- > Acceleration due to gravity (g): 9.8 m/s^2 . Decreases with altitude and depth. Maximum at poles.
- > Weight = mg (varies with location). Mass is constant everywhere.
- > Escape Velocity: Minimum velocity to escape Earth's gravity = 11.2 km/s .

PHYSICS - 3. ENERGY, WORK & POWER

- > Work = Force x Displacement x $\cos(\theta)$. Unit: Joule.
- > Kinetic Energy = $\frac{1}{2}mv^2$. Potential Energy = mgh .
- > Power = Work/Time. Unit: Watt. 1 HP = 746 Watts.
- > Conservation of Energy: Energy can neither be created nor destroyed, only transformed.
- > Types of energy: Mechanical, Thermal, Chemical, Electrical, Nuclear, Solar, Wind.

PHYSICS - 4. HEAT & THERMODYNAMICS

- > Temperature scales: Celsius (C), Fahrenheit (F), Kelvin (K). $C/5 = (F-32)/9$. $K = C + 273$.
- > Absolute Zero: $0\text{ K} = -273.15\text{ C}$. Lowest possible temperature.
- > Heat transfer: Conduction (solids), Convection (fluids), Radiation (no medium needed).
- > Specific Heat: Amount of heat to raise 1 kg by 1 C. Water has highest (4186 J/kg/K).
- > Latent Heat of Fusion (ice): 334 J/g. Latent Heat of Vaporization (water): 2260 J/g.
- > Thermometers: Clinical (35-42 C), Lab (uses mercury/alcohol).

PHYSICS - 5. LIGHT & OPTICS

- > Speed of light: 3×10^8 m/s (in vacuum). Light travels fastest in vacuum, slowest in denser medium.
- > Reflection: Angle of incidence = Angle of reflection. Mirror formula: $1/f = 1/v + 1/u$.
- > Refraction: Bending of light when passing from one medium to another. Snell's Law.
- > Total Internal Reflection: When light passes from denser to rarer medium at angle $>$ critical angle. Used in optical fibers.
- > Concave mirror: Converging. Used in torches, headlights, solar cookers, shaving mirrors.
- > Convex mirror: Diverging. Used in rear-view mirrors (wider field of view).
- > Convex lens: Converging. Used in magnifying glass, camera, projector, spectacles for hypermetropia.
- > Concave lens: Diverging. Used in spectacles for myopia (short-sightedness).
- > Dispersion: Splitting of white light into 7 colors (VIBGYOR). Rainbow is natural dispersion.
- > Scattering: Sky appears blue (shorter wavelength scattered more - Rayleigh scattering). Sunset appears red.

PHYSICS - 6. SOUND

- > Sound is a mechanical wave (needs medium). Cannot travel in vacuum.
- > Speed: Solids > Liquids > Gases. In air at 20°C: ~343 m/s.
- > Frequency: Hertz (Hz). Human hearing range: 20 Hz - 20,000 Hz.
- > Infrasound: < 20 Hz (elephants, earthquakes). Ultrasound: > 20,000 Hz (bats, SONAR, medical imaging).
- > Echo: Reflected sound. Minimum distance for echo: 17.2 m.
- > SONAR: Sound Navigation And Ranging. Used to detect underwater objects.
- > Doppler Effect: Change in frequency due to relative motion of source and observer.
- > Loudness measured in Decibels (dB). Normal conversation: 60 dB. Pain threshold: 120 dB.

PHYSICS - 7. ELECTRICITY & MAGNETISM

- > Current (I) = Charge/Time. Unit: Ampere (A). Flow of electrons.
- > Voltage (V) = Work/Charge. Unit: Volt. EMF of battery drives current.
- > Resistance (R) = V/I (Ohm's Law). Unit: Ohm. Depends on length, area, material, temperature.
- > Power (P) = $VI = I^2R = V^2/R$. Unit: Watt. 1 Unit (kWh) = 1000 W for 1 hour.
- > Series circuit: Same current, voltage divides. $R_{\text{total}} = R_1 + R_2 + R_3$.
- > Parallel circuit: Same voltage, current divides. $1/R_{\text{total}} = 1/R_1 + 1/R_2 + 1/R_3$.
- > Fuse: Safety device. Melts and breaks circuit on overload. Made of alloy of tin and lead.
- > Earthing: Connecting electrical devices to ground for safety.
- > Electromagnet: Magnet created by electric current. Used in motors, generators, bells, cranes.
- > Transformer: Changes voltage (step-up/step-down). Works on AC only. Principle: electromagnetic induction.
- > Generator: Converts mechanical energy to electrical. Motor: Converts electrical to mechanical.

CHEMISTRY - 1. ATOMIC STRUCTURE & PERIODIC TABLE

- > Atom: Smallest unit of element. Nucleus (protons + neutrons) + Electrons in orbits.
- > Atomic Number (Z) = Number of protons. Mass Number (A) = Protons + Neutrons.
- > Isotopes: Same atomic number, different mass number. E.g., H-1, H-2 (Deuterium), H-3 (Tritium).
- > Isobars: Same mass number, different atomic number. E.g., Ca-40 and Ar-40.
- > Modern Periodic Table: 118 elements, 7 periods, 18 groups. Based on atomic number.
- > Groups: Vertical columns. Periods: Horizontal rows.
- > Metals: Left side. Good conductors, lustrous, malleable, ductile. Lose electrons.
- > Non-metals: Right side. Poor conductors, brittle. Gain electrons.
- > Metalloids: Properties of both (Si, Ge, As, Sb, Te, Po).
- > Noble Gases (Group 18): Helium, Neon, Argon, Krypton, Xenon, Radon. Stable, inert.

CHEMISTRY - 2. ACIDS, BASES & SALTS

- > Acids: Sour taste, turn blue litmus red, $\text{pH} < 7$. Release H^+ ions in water.
- > Bases: Bitter taste, soapy, turn red litmus blue, $\text{pH} > 7$. Release OH^- ions.
- > pH Scale: 0-14. 7 = neutral. < 7 = acidic. > 7 = basic.
- > Strong Acids: HCl , H_2SO_4 , HNO_3 . Weak Acids: CH_3COOH (vinegar), H_2CO_3 .
- > Strong Bases: NaOH , KOH . Weak Bases: NH_4OH , $\text{Mg}(\text{OH})_2$.
- > Neutralization: Acid + Base = Salt + Water.
- > Common Salts: NaCl (table salt), NaHCO_3 (baking soda), Na_2CO_3 (washing soda), CaOCl_2 (bleaching powder), $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (Plaster of Paris).
- > Indicators: Litmus, Phenolphthalein, Methyl Orange, Universal Indicator.

CHEMISTRY - 3. METALS & NON-METALS

- > Most abundant metal in Earth's crust: Aluminium. Most abundant element: Oxygen.
- > Hardest natural substance: Diamond (carbon). Hardest metal: Chromium.
- > Most ductile metal: Gold. Most malleable: Gold. Best conductor of heat/electricity: Silver.
- > Lightest metal: Lithium. Heaviest naturally occurring: Uranium. Liquid metal: Mercury (Hg).
- > Alloys: Bronze (Cu+Sn), Brass (Cu+Zn), Steel (Fe+C), Stainless Steel (Fe+Cr+Ni+C), Solder (Pb+Sn).
- > Reactivity Series: $K > Na > Ca > Mg > Al > Zn > Fe > Ni > Sn > Pb > H > Cu > Hg > Ag > Au > Pt$.
- > Corrosion: Rusting of iron ($Fe_2O_3 \cdot xH_2O$). Prevention: painting, galvanizing, oiling, alloying.
- > Non-metals that are good conductors: Graphite (carbon). Liquid non-metal: Bromine.

CHEMISTRY - 4. CARBON & ITS COMPOUNDS

- > Carbon has 4 valence electrons. Forms covalent bonds. Shows catenation (long chains).
- > Allotropes of Carbon: Diamond, Graphite, Fullerene (C₆₀), Graphene, Carbon Nanotubes.
- > Diamond: Hardest, insulator, transparent. Graphite: Soft, conductor, lubricant.
- > Hydrocarbons: Methane (CH₄), Ethane (C₂H₆), Propane (C₃H₈), Butane (C₄H₁₀).
- > Alkanes (C_nH_{2n+2}): Single bonds. Alkenes (C_nH_{2n}): Double bonds. Alkynes (C_nH_{2n-2}): Triple bonds.
- > Ethanol (C₂H₅OH): Drinking alcohol. Methanol (CH₃OH): Poisonous. Ethanoic Acid (CH₃COOH): Vinegar.
- > Soaps: Sodium/Potassium salts of fatty acids. Detergents: Synthetic, work in hard water.

BIOLOGY - 1. CELL BIOLOGY

- > Cell: Basic structural and functional unit of life. Discovered by Robert Hooke (1665).
- > Prokaryotic: No membrane-bound nucleus (bacteria). Eukaryotic: Has nucleus (plants, animals).
- > Cell Organelles: Nucleus (DNA), Mitochondria (powerhouse), Ribosome (protein synthesis), ER (transport), Golgi (packaging), Lysosome (suicide bags), Chloroplast (photosynthesis - plants only).
- > Cell Division: Mitosis (2 identical cells, growth/repair) and Meiosis (4 cells with half chromosomes, gamete formation).
- > DNA: Deoxyribonucleic acid. Double helix. Watson & Crick (1953). Carries genetic info.
- > RNA: Single-stranded. Types: mRNA, tRNA, rRNA. Helps in protein synthesis.
- > Human cells have 46 chromosomes (23 pairs). Sex chromosomes: XX (female), XY (male).

BIOLOGY - 2. HUMAN BODY SYSTEMS

- > Digestive System: Mouth > Esophagus > Stomach (HCl, pepsin) > Small intestine (absorption) > Large intestine > Rectum.
- > Liver: Largest internal organ. Produces bile. Detoxification. Stores glycogen.
- > Respiratory System: Nose > Pharynx > Larynx > Trachea > Bronchi > Lungs (alveoli - gas exchange).
- > Circulatory System: Heart (4 chambers), Blood vessels. Blood groups: A, B, AB (universal recipient), O (universal donor).
- > Heart: Right side = deoxygenated blood. Left side = oxygenated blood. Double circulation.
- > Blood Components: RBC (carry O₂, no nucleus), WBC (immunity), Platelets (clotting), Plasma (55% of blood).
- > Nervous System: Brain + Spinal Cord + Nerves. Brain parts: Cerebrum (thinking), Cerebellum (balance), Medulla (involuntary actions).
- > Endocrine System: Ductless glands. Pituitary (master gland), Thyroid (metabolism), Adrenal (fight/flight), Pancreas (insulin), Pineal (melatonin).
- > Excretory System: Kidneys filter blood. Nephron: functional unit. Produces urine.
- > Skeletal System: 206 bones in adult. Smallest: Stapes (ear). Largest: Femur (thigh).
- > Muscular System: 639 muscles. Largest: Gluteus Maximus. Smallest: Stapedius.

BIOLOGY - 3. DISEASES

Bacterial Diseases:

- > Tuberculosis (Mycobacterium), Cholera (Vibrio cholerae), Typhoid (Salmonella typhi), Plague (Yersinia pestis), Tetanus (Clostridium tetani), Leprosy (Mycobacterium leprae), Pneumonia.

Viral Diseases:

- > COVID-19 (SARS-CoV-2), Dengue (Aedes mosquito), Rabies, Polio, HIV/AIDS, Influenza, Measles, Hepatitis, Ebola, Nipah, Chickenpox, Mumps.

Protozoan Diseases:

- > Malaria (Plasmodium, Anopheles mosquito), Amoebic Dysentery, Sleeping Sickness (Tse-tse fly), Kala-azar (Sandfly).

Fungal Diseases:

- > Ringworm, Athlete's Foot, Candidiasis.

Deficiency Diseases:

- > Scurvy (Vitamin C), Rickets (Vitamin D), Beriberi (Vitamin B1), Pellagra (Vitamin B3), Night Blindness (Vitamin A), Anemia (Iron), Goitre (Iodine), Kwashiorkor (Protein), Marasmus (Protein+Calories).

BIOLOGY - 4. VITAMINS & NUTRITION

Vitamin	Chemical Name	Source	Deficiency
A	Retinol	Carrot, Liver, Fish oil	Night Blindness
B1	Thiamine	Whole grains, Meat	Beriberi
B2	Riboflavin	Milk, Eggs, Liver	Cheilosis
B3	Niacin	Meat, Fish, Peanuts	Pellagra
B6	Pyridoxine	Meat, Banana, Nuts	Anemia
B9	Folic Acid	Leafy greens, Beans	Megaloblastic anemia
B12	Cobalamin	Meat, Fish, Dairy	Pernicious anemia
C	Ascorbic Acid	Citrus fruits, Amla	Scurvy
D	Calciferol	Sunlight, Fish oil	Rickets (children)
E	Tocopherol	Vegetable oils, Nuts	Infertility
K	Phylloquinone	Green leafy veg	Bleeding disorder

BIOLOGY - 5. ECOLOGY & ENVIRONMENT

- > Ecosystem: Biotic (living) + Abiotic (non-living) components interacting together.
- > Food Chain: Producer > Primary Consumer > Secondary Consumer > Tertiary Consumer > Decomposer.
- > Biodiversity Hotspots in India: Western Ghats, Eastern Himalayas, Indo-Burma, Sundaland (Nicobar).
- > Biosphere Reserves in India: 18 total. Nilgiri (first, 1986), Pachmarhi, Sundarbans, Gulf of Mannar, etc.
- > National Parks: 106 in India. Largest: Hemis (Ladakh). Oldest: Jim Corbett (1936, Uttarakhand).
- > Wildlife Sanctuaries: 567 in India. Tiger Reserves: 58 (Project Tiger started 1973).
- > Greenhouse Gases: CO₂, CH₄, N₂O, CFCs. Cause global warming.
- > Ozone Layer: In stratosphere. Protects from UV radiation. CFCs cause depletion.
- > Ramsar Sites (Wetlands): India has 80+ Ramsar sites. Largest: Sundarbans.